A case study of the time and nature of paleomagnetic resetting in a mafic complex in New England

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It appears that a hydrothermal alteration and development of magnetite is largely responsible for a paleomagnetic resetting in a belt of mafic–intermediate rocks in eastern Massachusetts. The age of this thermal and metasomatic event is found to be about 370 m.y. by both geochronological and paleomagnetic evidence. This age does not coincide with the emplacement of plutons within the region (600 and 450 m.y.), nor with the 340 m.y. uplift and cooling shown by K–Ar ages on biotites.

The age and tentative pole position obtained favour the hypothesis that an eastern region in New England and Maritime Canada comprised a single block in the Devonian that was separate from North America at that time, and fits the polar wanderpath of North Africa.

Il semble qu'une altération hydrothermale et une apparition de magnétite soient en grande partie responsable de la réorientation paléomagnétique dans une ceinture de roches moyennement basiques de l'est du Massachusetts. L'âge de cet événement thermal et métasomatique est d'environ 370 Ma selon les méthodes paléomagnétique et géochronologique. Cet âge ne coïncide pas avec la mise en place des plutons de la région (600 et 450 Ma) non plus qu'avec le soulèvement d'il y a 340 Ma et le refroidissement indiqués par les âges du K–Ar des biotites.

L'âge et l'emplacement suggéré du pôle que nous avons obtenu favorisent l'hypothèse qu'une portion de l'est de la Nouvelle Angleterre et des Maritimes canadiennes formaient durant le Dévonien une seule masse qui était isolée de l'Amérique du Nord à ce moment là, et coïncide avec la trajectoire du pôle de l'Afrique du Nord.

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Introduction

Recently there has been increasing interest in the use of metamorphic rocks for paleomagnetic studies in the Precambrian (Buchan and Dunlop 1973; Irving *et al.* 1974; Fahrig *et al.* 1974; Poorter 1975; McWilliams and Dunlop 1975). One of the problems has been the dating of the resetting event, and it has been a matter of debate whether the K–Ar age of biotite may be used to give an indication of this time. However, the K–Ar ages of hornblende are often several tens of millions of years greater than those of biotite in the same system, so that the choice is clearly arbitrary unless other evidence is available. In this study there is such evidence, due to a multiplicity of datable events. Fortunately, the resetting is young enough so that fine distinctions in the dating can be used.

It is to be clearly understood that in this investigation we have not intended to establish a paleopole, with full commitment to that task alone, but rather to have examined the timing and nature of resetting. Only 22 independent samples (one per site in most cases) have been used, from which the 206 specimens have been cut. The statistical parameters are representative of the scatter of the 22 samples and not of the specimens taken within each site.

Geologic Setting

Mafic Complex

Figure 1 shows an enclave of mafic to intermediate, volcanic–hypabyssal rocks within an igneous–metamorphic complex in eastern Massachusetts. It is bounded by the Andover granite, and gneisses and other metamorphic rocks of

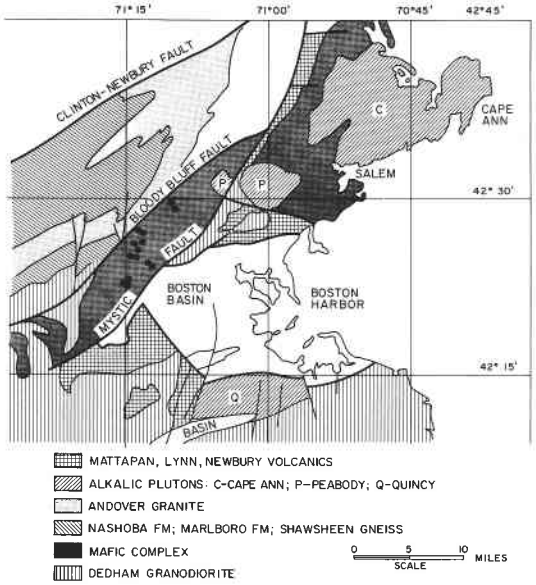


FIG. 1. Igneous–metamorphic complex in Eastern Massachusetts showing principal units only. The undifferentiated volcanic–plutonic region (dark-shaded) includes a late Precambrian mafic volcanic–hypabyssal suite mostly altered to greenschist grade, and some intrusives such as the Salem gabbrodiorite, which may be early Paleozoic in age. It is intruded by the Dedham granodiorite (600 m.y.) in the southern part of the sheet, and by the Cape Ann and Peabody granites (450 m.y.) in the northern part.

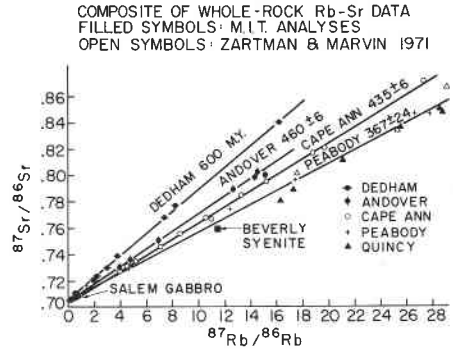


FIG. 2. Plots of Rb–Sr whole-rock data for the Dedham granodiorite, the Andover, Cape Ann, Peabody, and Quincy granites, and the Salem gabbro. The alkaline Cape Ann, Peabody, and Quincy granites are Ordovician (450 m.y.) in age, but variously affected by metasomatic activity during the Acadian orogeny (about 360 m.y.).

the Nashoba and Marlboro Formation on the northwest with a fault contact; and has been invaded by the Cape Ann and Peabody granites on the north, and by the Dedham granodiorite on the south. The rocks within the enclave include gabbro to quartz-diorite intrusives, mafic migmatites, and flows, but has not been divided in sufficient detail to indicate various units on the map. For this reason we will refer to it as the Mafic Complex.

Age data on the younger invading plutons are critical to this study and must be considered in detail if the age of the magnetic resetting is to be established. Also of importance is the recognition of the major faulting, across which there is an abrupt change in lithology and metamorphism in the area of interest. In particular, the biotite gneisses and schists northwest of the Bloody Bluff fault have no counterpart on the southwest side.

In the area of the sample sites within the Mafic Complex, the rocks are found to be altered to various degrees: always with development of epidote and magnetite, and generally with sericite in the plagioclase. Most of the rocks contain biotite, and over half have hornblende. The plagioclase is commonly in the range An_{25-45}; occasionally An_{65-70}. Much of it has probably been albitized, with CaO going into epidote where it has not been replaced by fine grained muscovite. The opaques are commonly distributed in close relationship with the alteration products, and for other reasons (discussed later) are believed to be largely secondary in origin. The area has been invaded to a varying extent by lighter-colored intrusive masses ranging from metres in width to aplite stringers. The variable hydrothermal alteration involved introduction of H_2O and K_2O, and produced a greenschist metamorphism mostly within the biotite isograd.

Dedham Granodiorite

The principal plutonic mass in the southern part of Fig. 1, the Dedham granodiorite, is dated by whole rock Rb–Sr analyses at about 600 m.y. if we include the earlier (Fairbairn *et al.* 1967*a*,*b*; and later unpublished) analyses made in this laboratory (see Fig. 2). There seems to be little doubt about this age, except that there may be some younger intrusive activity in the region locally. The Dedham granodiorite invades the Mafic Complex, so that the latter is, in part at least, Precambrian in age.

Andover Granite

This unit shows the typical late Ordovician age (460 m.y., Fig. 2) of other plutons in the re-

TABLE 1. Age determinations

Location or rock unit			References
A. *K–Ar Mineral Ages*	Biotite	Amphibole	
(1) Mafic Complex from which paleomagnetic samples were taken:			
Trapelo Road, Lincoln	323	386	This study
Route 2, Belmont Hill	359	380	This study
Route 3, Lexington	346	342	This study
Childs Street, Lexington	314	336	This study
Belfrey Hill, Lexington	342	399	This study
(2) Alkaline granites			
Cape Ann	382	380	Zartman and Marvin 1971
		393	Zartman and Marvin 1971
		397	Zartman and Marvin 1971
		374	Zartman and Marvin 1971
		390	Zartman and Marvin 1971
Peabody	367	370	Zartman and Marvin 1971
		403	Zartman and Marvin 1971
		384	Zartman and Marvin 1971
		370	Zartman and Marvin 1971
		350	Zartman and Marvin 1971
		367	Zartman and Marvin 1971
Average	340	380	
B. *Rb–Sr and Pb–U Ages*	Whole-Rock Rb–Sr	Zircon Pb–U	
(1) Younger Invading Plutons:			
Cape Ann	435	450	Zartman and Marvin 1971
Peabody	(367)	440	Zartman and Marvin 1971
Quincy		440	Zartman and Marvin 1971
Andover	460		Fairbairn *et al.* 1967
Ayer	460		Heath 1965
Average	450		
(2) Older Invading Plutons:			
Dedham Granodiorite	591		Fairbairn *et al.* 1967
Dedham Gabbrodiorite	605		A. Kovach, personal communication 1975

gion (Handford *et al.* 1965). It has clearly intruded a group of rocks on the northwest side of the Bloody Bluff fault that has been considered to be part of the Mafic Complex (not shown in Fig. 1). Although of approximately the same age as the alkalic plutons to the east of the fault, the Andover granite differs from them markedly in composition, and shows none of the effects of the alkalic metasomatism discussed below.

Cape Ann Granite

This alkalic intrusion is believed to be genetically related to the Peabody and Quincy intrusions on the basis of composition. In the group as a whole, the principal feldspar is a microperthitic intergrowth of microcline and sodic plagioclase; ferrohornblende, riebeckite, aegerine–augite, and more rarely, biotite and fayalite are characteristic ferromagnesians. The rock is massive and undeformed, and apparently intrudes all surrounding units. Radiometric age investigations by Zartman and Marvin (1971) disclosed an age of 435 $\pm$ 6 m.y. by whole-rock Rb–Sr (reproduced in Fig. 2), which was nearly concordant with the zircon Pb–U age of about 450 m.y. On the other hand, they found hornblende K–Ar ages to average 387 m.y. (Table 1) and a biotite K–Ar age of 350 m.y.

Peabody Granite

There are several bodies of this light to dark green, coarse grained ferrohornblende granite intruding the Mafic Complex. Zartman and Marvin (1971) have found a decidedly lower Rb–Sr whole-rock apparent age (see Fig. 2: 367 $\pm$ 24 m.y.) for this rock, but also found a zircon Pb–U age of about 440 m.y. and an average hornblende K–Ar age of 375 m.y.

Quincy Granite

This granite is the most complex of all granites studied radiometrically, to our knowledge. Bottino *et al.* (1970) found an alignment of Rb–Sr whole-rock points at 365 ± 7 m.y., but a scatter of other points for rhyolitic and porphyritic phases that did not make sense. Zartman and Marvin (1971) found a Rb–Sr whole-rock value of 313 ± 22 m.y., but a zircon Pb–U value of about 440 m.y., and riebeckite K–Ar ages averaging 440 m.y.

Metamorphic Age Relationships

Because the alkalic Cape Ann, Peabody, and Quincy granites are similar compositionally, and all show similar zircon ages of 440 m.y. or slightly greater, and the Cape Ann Rb–Sr and the Quincy riebeckite K–Ar age are both about 440 m.y., it was concluded by Zartman and Marvin that the original emplacement of these alkalic plutons was of that age. However, east of the Bloody Bluff fault, it seems that there was an extensive thermal and associated metasomatic episode in the interval of about 340–400 m.y., related to the Early and Middle Devonian Acadian orogeny, that was severe enough to affect some of the whole-rock Rb–Sr and the mineral ages given in Fig. 2 and Table 1.

Within the Mafic Complex, the patchwork of hydrothermal alterations seems to be related to the abundance of stringer stockworks and apophyses of silicic rocks. These are also evident locally within the alkalic plutons, so it is logical to attribute the alterations in the Mafic Complex to the same thermal and hydrothermal metasomatic episode that reset the Rb–Sr ages of the Peabody and Quincy granites (about 365 m.y.).

The actual determination of the age of these alterations is based on direct measurements on biotite and amphibole. Table 1 gives a list of K–Ar age determinations on samples of biotite and hornblende obtained within the area sampled. The biotite values represent cooling ages. There is general agreement with the K–Ar data on the alkalic plutons to the north (Table 1).

If we assume the entire belt to be rising with a constant thermal gradient (due to constant uplift rate and heat flux) and use blocking temperatures for Ar in biotite and amphibole of 300 and 500 °C, respectively, we have the relationship shown in Fig. 3. Uplift rate, biotite blocking depth, and amphibole blocking depth are shown

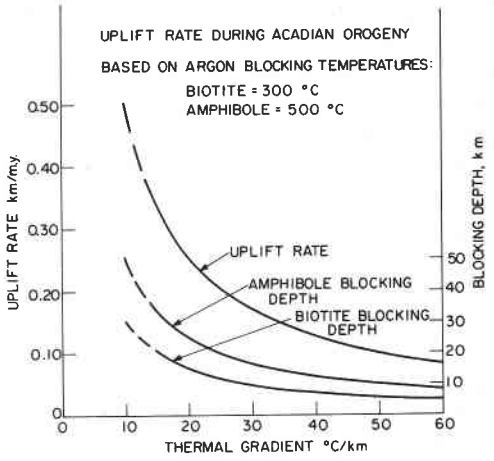


FIG. 3. Schematic relationships between uplift rate during the Acadian orogeny and depths at which argon was retained by biotite and hornblende, as a function of the thermal gradient.

as functions of thermal gradient. The range of temperatures estimated for the greenschist grade of metamorphism is from 300 to 500 °C (Winkler 1967), which in Fig. 3 just spans the biotite–amphibole Ar-blocking depth range. Thus, we have several choices regarding the time of resetting of magnetic remanence in these rocks (to be discussed below):

(a) the magnetic resetting was due to the *thermal* pulse at a temperature close to that of the Ar-blocking of hornblende, *i.e.* about 500 °C, which was at 380 m.y.

(b) the magnetite resetting was due to either recrystallization or new development of magnetite at a temperature in the center of the greenschist range (*i.e.* about 400 °C), which would be at about 360 m.y.

(c) the magnetite was developed along with epidote and chlorite by hydrothermal alteration at the time of the metasomatic Rb–Sr age resetting of the Peabody granite, namely at 365 m.y.

We have used these arguments and data, together with the paleomagnetic evidence presented below, to place the time of resetting of magnetic remanence at 370 ± 10 m.y. It seems unlikely that the hydrothermal activity continued until cooling and uplift reached the Ar-blocking temperature of biotite (340 m.y.). Clearly the alterations giving rise to varied amounts of epidote, chlorite, magnetite, and muscovite were dependent on the availability of H_2O and K_2O and not solely on temperatures. This suggests

a peak of activity during the intrusion of the aplites. Whether the biotite and hornblende developed at this time or earlier is still open to question, but we are inclined to place them along with other alterations involving H_2O and K_2O as mobile components and suggest that the temperature varied throughout the entire system because of the locally transported heat.

Paleomagnetic Measurements

Twenty-two oriented samples for paleomagnetic investigation were taken from 21 sites over a 20 km stretch within the Mafic Complex. Sampling areas are shown in Fig. 1 with more than one site per area in some cases. These supplied a total of 206 specimens. Magnetic data were obtained using a Schonstedt spinner magnetometer. Step demagnetization was carried out for each sample, mostly up to 600 Oe. The samples showing a minimum orientation of fabric were selected, to minimize effects of anisotropy. Anisotropy measurements indicated that this is not a significant problem. Fisher (1953) statistical methods were applied at the between-site level, a 'site' being here defined as a physical locality, generally represented by a single independently oriented sample. An average paleopole position for 24 specimens of the Upper Triassic Palisade Sill agreed well with the established pole position for that unit, and furnished an interlaboratory test of field and laboratory procedures (McElhinny 1973).

Magnetite occurs primarily as 'dusty' grains within, or small anhedral grains at the boundaries of, the alteration products in the rocks. The anhedral grains range up to 1000 μm and the dusty particles up to 100 μm, in width. Only rarely were large, subhedral, apparently primary, magnetite grains observed. These are partly resorbed, suggesting that small primary grains have been entirely consumed during metamorphic recrystallization.

A considerable amount of time and effort was spent in an attempt to use Transmission Electron Microscopy to determine the structure of the larger magnetite grains. It was found that some of the larger grains were actually clusters of smaller grains, based on a differentiation of the orientations of their crystal structure, but we were not able to apply the technique to the ultimate extent of its capability: namely, to determine the domain state of the magnetite. A TEM study of the grains from a few of the samples indicates that the dusty magnetite grains are homogeneous and range in size down to 100 Å (1 Å = 0.1 nm). In the anhedral grains, the agglomerations consist of smaller particles ranging from 100 Å to several micrometres in size. Both the dusty component and the agglomerations, therefore, appear to contain superparamagnetic, single domain and multidomain grains. It is plausible (probable) that a large portion of the stable remanence in the samples is carried by single-domain-like particles.

Calculated values of Q_t (the Koenigsberger ratio; Stacey 1967) were generally of the order of 1–10, with some samples giving values as high as 30, supporting the suggestion that single-domain behavior dominates the remanence. The Lowrie–Fuller (1971) test, comparing the demagnetization behavior of natural remanence (NRM) with that of saturation remanence (IRM_s), was less conclusive: of three specimens tested, one clearly indicated single-domain magnetites, while two showed only marginally single-domain behavior.

The magnetite content of the samples varied from 0.1% to 10% by volume, and in individual specimens is in good agreement with estimates from bulk susceptibility measurements, using either the relationship of Balsley and Buddington (1958) or that of Mooney and Bleifuss (1953). Sample NRM intensities ranged between 10^{-2} and 10^{-4} emu/cm^3, while 'cleaned' values ranged between 10^{-3} and 10^{-5} emu/cm^3. The sample cleaning level was chosen (*e.g.* Fig. 4) as that level at which the best within-sample clustering of specimen directions was obtained (minimum α_{95} and maximum k; Fisher 1953).

Table 2 lists the averaged paleopoles and relevant Fisher statistical parameters corresponding to the cleaned magnetic directions (Cox and Doell 1960).

Behavior of different samples during AF demagnetization varied considerably, as illustrated in Fig. 4. Most rocks were characterized by a low median destructive field (below 100 Oe; Table 2 and Fig. 4, Rock no. 1), but in some the NRM intensity was more stable (Fig. 4, Rock 19), and a few showed median destructive fields of 500 Oe or higher (Fig. 4, Rock 6). The generally low mean destructive fields may be explained by the predominance of low coercivity, multidomain particles in the samples. However, the existence

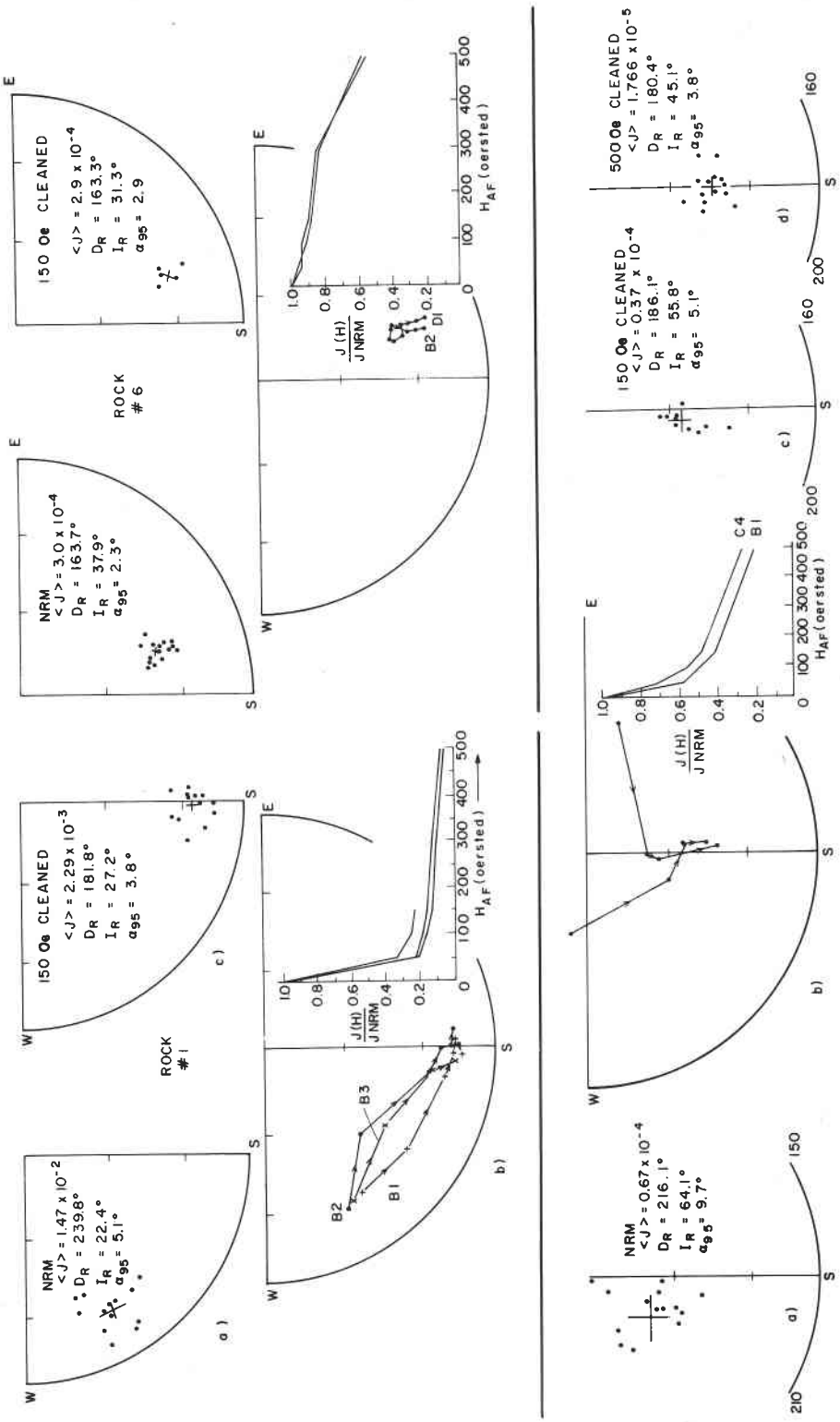


FIG. 4. AF cleaning data for samples 6, 19, and 1, showing cleaning behavior of samples varying from higher to lower mean destructive fields.

TABLE 2. Summary of paleomagnetic data

Sample No.	No. of Specimens	NRM Average declination	NRM Average inclination	Selected AF cleaning level (Oe)	Median destructive field (Oe)	Cleaned average declination	Cleaned average inclination
1	13	239.8	22.4	150	45	181.8	27.2
2	14	192.8	51.5	150	45	193.9	35.4
6	10	163.7	37.9	500	500	160.0	24.4
7	10	356.3	40.7	500	100	161.0	58.9
8	6	344.6	64.0	700	100	168.2	54.2
16	10	35.6	78.6	$\gg$500	40	137.5	4.3
17	10	189.4	39.2	300	150	176.9	28.5
18	6	279.5	69.4	100	150	143.6	52.9
19	14	216.1	64.1	500	125	180.4	45.1
20	6	228.0	11.0	500	30	170.0	46.0
21	10	309.8	82.5	150	30	141.8	55.1
23	8	122.6	−4.5	500	30	178.9	14.6
24	7	329.4	54.2	700	40	175.2	55.6
71	7	135.6	29.5	50	35	144.0	38.3
72	6	78.3	58.8	100	70	144.9	60.1
77	9	163.2	34.1	100	200	146.0	31.1
78	5	159.7	16.8	200	500	151.2	4.7
79	11	81.3	68.4	100	300	134.7	66.8
81	11	205.9	54.6	100	150	174.6	29.7
88	9	172.6	47.9	50	100	183.7	42.0
8218	9	51.9	34.5	500	40	167.1	66.1
8220	15	(314.8)*	(−27.2)	150	40	(321.1)*	(−23.4)*
Mean for 22 sites:		179.4	70.0			162.7	39.3

Corresponding VGP: P lat. = −23°, P long. = −54°, K = 11, and α_{95} = 10° for the 22-site mean.
*The only sample with normal polarity; all others reversed.

of a higher-coercivity fraction of grains capable of retaining remanence over a long geologic period of time is implied by the reasonably tight grouping of cleaned directions. The majority of samples showed a positive correlation among several properties: (1) higher stability of NRM intensity; (2) a smaller departure from NRM direction during magnetic cleaning; and (3) greater proportion of dispersed dusty grains relative to clusters of grains, implying that the dusty fraction contains a higher precentage of high coercivity material.

The average paleopole position for the 22 samples and its α_{95} confidence limit are plotted together with selected North American polar wander curves from McElhinny (1973) and McElhinny and Opdyke (1973) in Fig. 5a and b, respectively. Regardless of which curve is preferred, our average pole position most closely corresponds to the Devonian pole position on the North American path, though the Mafic Complex is known to be considerably greater in age, and the intruding granites to be Ordovician. A discussion of the North American Devonian pole position follows later.

Thus we have the additional paleomagnetic evidence that the epidote–chlorite–magnetite alterations took place during the 370 ± 10 m.y. Devonian thermal episode, and not during the 450 m.y. emplacements of the Ordovician plutons, nor during the 335 m.y. Carboniferous cooling and uplift time shown by biotite K–Ar ages.

Global Paleomagnetic Considerations

The proximity of our pole to the McElhinny (1973) and McElhinny and Opdyke (1973) Devonian averages for North America is possibly of significance to the concept of North American and African continental drift. Both of these average poles are based on results from the northern Maine – southwestern New Brunswick Perry Formation (Black 1964; Phillips and Heroy 1966; Robertson *et al.* 1968). (The McElhinny (1973) average also includes the Clam Bank Formation results from Newfoundland, which Black (1964) claims may have suffered Devonian rotation with respect to Maritime Canada). The coastal area between southern New England and New Brunswick seems, there-

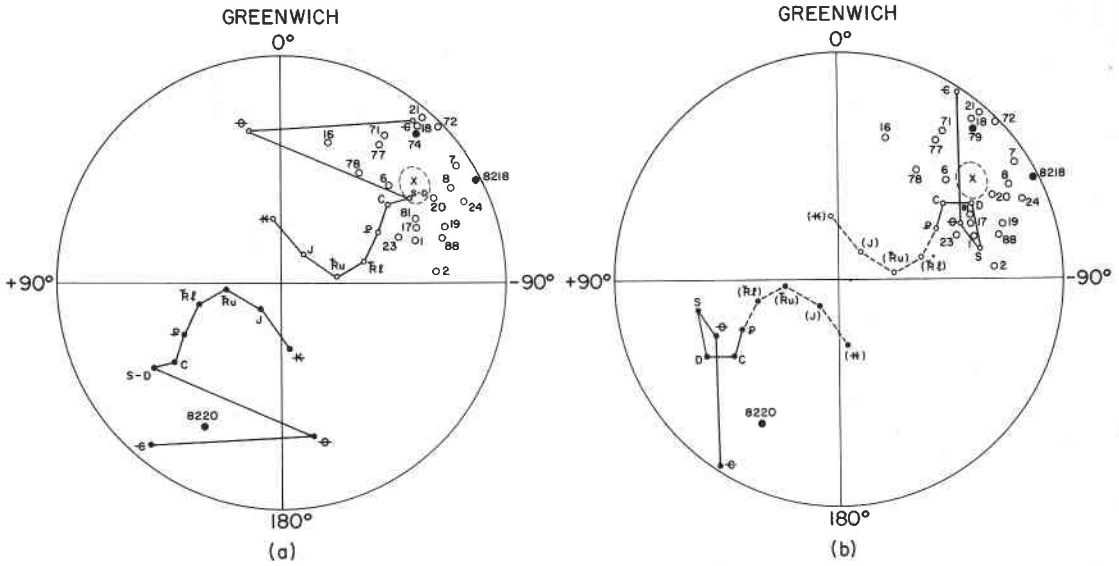


FIG. 5. VGP positions for different sites, and the average paleopole position plotted as x for all the samples. The dashed α_{95} confidence limit for this average point is based principally on the variation between specimens from each sample and not between samples at each site, because there were only 22 independent samples taken from the 19 sites. This average point is compared with North American polar wander paths from McElhinny (1973) in 5(a), and from McElhinny and Opdyke (1973) in 5(b). Points with open circles are in the southern hemisphere and with filled circles in the northern. All sites show reversed polarity except No. 8220, which is normal.

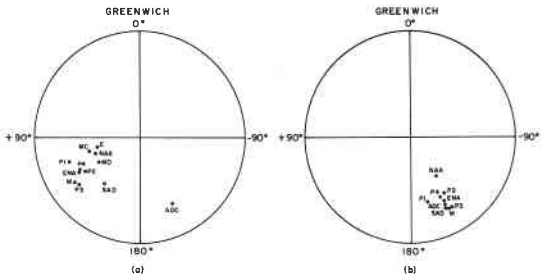


FIG. 6. Comparison of Massachusetts paleopole with Devonian poles for Maine – New Brunswick and Newfoundland. In Fig. 6(a), these are compared with other North American poles and a South American average Devonian pole. In Fig. 6(b), the various Devonian poles for North and South America are compared with the Devonian–Carboniferous African Dwyka varves pole after rotation into a reconstructed Pangaea. The northeastern block appears to fit the Africa – South America block better than the North America block, suggesting that it was part of Africa prior to the closure of the proto-Atlantic. Notation: ADC, average Dwyka varves; MC, Columbus limestone; MO, Ohio shale; E, Arizona Martin formation; P1-P4, Perry formation; M, Massachusetts greenstone; NAA, North American average; ENA, eastern North American average; SAD, South American Devonian average.

fore, to have been part of the same coherent block during the Devonian.

That this block was part of the Devonian North American mass is, however, questionable. The Massachusetts greenstone pole and the Perry poles are plotted in Fig. 6a, along with Devonian poles from the Ohio Columbus Limestone and Ohio Shale (Martin 1975) and the Arizona Martin Formation (Bressler and Elston 1974). The average of the latter three poles, all of which are from demonstrably cratonic areas, plots some 22° from the average of the Perry and Massachusetts poles. However, after rotating North America to Africa around the Bullard *et al.* (1965) center of rotation for the closing of the Atlantic, the Massachusetts–Perry average plots only 3° from the African Devonian–Carboniferous Dwyka Varves pole (McElhinny and Opdyke 1968; Fig. 6b).

As general continental drift theory attaches South America to Africa during the Devonian, an average South American pole for the Devonian (40° S, 38° W) was extrapolated from the data of McElhinny (1973; Fig. 6a) and was ro-

tated about the Bullard *et al.* (1965) South America to Africa center of rotation. The resultant South American Devonian pole position plots within 3° of the African Dwyka pole and within 6° of the rotated New England – New Brunswick pole (Fig. 6b).

These data seem to indicate that eastern New England and New Brunswick were part of Africa during the Late Devonian, and that Africa and North America had not come together by that time, as is commonly believed (Bird and Dewey 1970; Schenk 1971; Hurley *et al.* 1974). Admittedly the existing data are sparse, but the results seem to support the conclusion of Scott (1975), based on more extensive paleomagnetic data, that southeastern Atlantic Canada, eastern New England, Africa, and South America were part of a coherent Paleozoic block that did not come together with North America until post-Devonian, and perhaps Permian, time.

Conclusions

The development of a significant fraction of high-coercivity magnetite grains in a suite of hydrothermal alteration products is indicated by our combined geochronologic and paleomagnetic study of a mafic complex in New England. Thus, regions containing mafic rocks (extrusive and intrusive) that have suffered deformation and igneous intrusion in an orogenic belt may offer the opportunity for obtaining dated paleopoles, if these alterations can be found and dated, and demonstrated to be post-deformation by general coherence of results. Dating by biotite K–Ar is ill-advised; amphibole K–Ar may better reflect the thermal and metasomatic event. The age of the invading plutons may antedate the resetting; in this example, the time difference was 80 m.y.

The coincidence of our tentative Massachusetts Devonian pole with others from Maine and New Brunswick suggests that the eastern New England – southeastern Canada area was part of the same coherent Paleozoic block. Paleomagnetic correlations further suggest that this block was part of an African – South American mass separate from North America during the Devonian.

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